Bridging the Gap Between Theory and Practice in Adversarial Machine Learning: A Systematic Cross-Venue Analysis of 454 Papers (2022–2025)

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Abstract

Machine learning systems are being used more often in security-critical applications, but there is still a significant gap between academic research in adversarial machine learning and the needs for real-world deployment. This systematic review looks at 454 papers published in four major security venues (ACM CCS, IEEE S&P, NDSS, and USENIX Security) covering the years from 2022 to 2025. Using the analytical framework introduced by Apruzzese et al. in "Real Attackers Don't Compute Gradients," we evaluate how current research addresses the divide between theory and practice.

Our analysis shows that 94.7% of the papers do not test on deployed systems, 67.8% depend on gradient access that is rarely available in real situations, 80.4% assume query budgets that go beyond typical operational limits, and 63.2% use white-box threat models. We identify five areas that need more attention in relation to real-world deployment: specifying threat models, accessing gradients, validating on real systems, ensuring diversity beyond image classification, and considering economic and human factors. Our temporal analysis from 2022 to 2025 shows that progress in these areas is gradual rather than swift.

We also examine how these gaps manifest in the emerging domain of large language models and multimodal foundation models. Despite operating in discrete token spaces with fundamentally different attack surfaces, these systems exhibit the same theory-practice disconnect: impractical gradient-based attacks, unrealistic query budgets, evaluation instability, and insufficient real-world validation. The systemic deployment of foundation models, where single systems serve numerous downstream applications, amplifies the practical importance of closing these gaps. We end with suggestions for researchers, venues, practitioners, and funding agencies to promote more effective adversarial machine learning research.

Keywords: adversarial machine learning, theory-practice gap, security research, systematic review, threat modeling

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1 Introduction

Machine learning has become an essential part of software systems in many areas. It is used in facial recognition at border crossings, fraud detection in financial services, navigation for autonomous vehicles, and content moderation on social platforms. Large language models now power customer service chatbots, code generation tools, and medical assistants. ML models now influence decisions that impact millions of people every day. This widespread use has led to greater scrutiny of vulnerabilities in these models. Researchers design specific inputs to trigger failures that system creators did not foresee.

The field of adversarial machine learning (AML) has emerged to explore these weaknesses and develop defenses. Over the last ten years, researchers have shown how sophisticated attacks can work: small changes that lead to image misclassification, poisoned training data that creates hidden backdoors, and privacy attacks that reveal sensitive information through targeted queries. In response, the community has proposed many defenses to improve model strength.

Despite these developments, a key question remains: **how well do the assumptions in academic research match the realities of real-world systems?**

In 2022, Apruzzese and colleagues published a thorough analysis titled "Real Attackers Don't Compute Gradients" [1]. They documented a gap between academic AML research and practical security needs. Their main finding was that academic studies often assume attackers can calculate gradients from target models, which requires access to the model's structure and parameters. However, real-world attackers usually lack this access. Often, simpler methods like social engineering or basic input manipulation are sufficient, making complex approaches unnecessary for them.

This systematic review asks: *how has the research community responded since this critique?*

To find answers, we analyzed 454 adversarial ML papers published from 2022 to 2025 in four leading security venues: ACM CCS (118 papers), IEEE S&P (79 papers), NDSS (49 papers), and USENIX Security (208 papers). We assessed each paper based on several factors related to practical relevance: realism of the threat model, computational needs, validation methods, and awareness of deployment challenges.

Our findings show that the gap between theory and practice is still significant. While the research community has made valuable contributions to our understanding of adversarial issues, the drive to publish often emphasizes innovation and theoretical strength over practical implementation and cost considerations. This review outlines the current state of the field, identifies common trends across venues, and provides practical recommendations for various stakeholders.

2 Background: Foundations of Adversarial Machine Learning

2.1 The Discovery of Adversarial Vulnerabilities

The modern study of adversarial machine learning began with the important work of Szegedy et al. [2]. They showed that small, barely noticeable changes to input data could lead state-of-the-art deep learning models to produce wrong outputs with high confidence. These changes were not just random mistakes; they were systematic vulnerabilities that could transfer between different models. Adversarial examples created for one model often worked against other models trained separately. This ability to transfer highlighted the nature of the vulnerability and sparked further research.

Goodfellow et al. [3] suggested that the linear behavior of neural networks, instead of their nonlinearity or overfitting, was the main cause of their vulnerability to adversarial attacks. They introduced the Fast Gradient Sign Method (FGSM), which made it easier to generate adversarial inputs using a single gradient step. The speed and effectiveness of FGSM set a standard for later research, as gradient-based attacks became the common approach.

These foundational studies shaped the direction of adversarial ML research in three significant ways. First, they assumed that adversaries had white-box access to models, including details about their architecture and gradient information. Second, they focused on imperceptible changes measured by specific distance metrics. Third, they highlighted optimization-based methods as the best way to generate adversarial inputs. While these frameworks allowed for thorough analysis, they also shifted research away from the practical realities of real-world attacks.

2.2 The Arms Race: Attacks and Defenses

The discovery of adversarial vulnerabilities triggered an ongoing cycle of proposed defenses and increasingly sophisticated attacks designed to circumvent them. This pattern has come to characterize adversarial machine learning research as an "arms race" between attackers and defenders [4]. Each new defense mechanism motivated adaptive attacks that exploited weaknesses in the defense strategy, which in turn motivated new defensive approaches. This adversarial co-evolution has shaped both the technical landscape and methodological standards of the field.

2.2.1 Advanced Gradient-Based Attacks

Building on the foundational work of Szegedy et al. [2] and Goodfellow et al. [3], researchers developed more sophisticated attack methods that optimized adversarial perturbations more precisely. A central challenge in designing adversarial attacks is balancing two competing objectives: maximizing the probability of misclassification while minimizing the perceptibility of the perturbation. To formalize this trade-off, researchers employ **distance metrics** that quantify how much an adversarial example differs from the original input.

The most commonly used distance metrics in adversarial ML are L_p norms, which measure the magnitude of perturbations applied to inputs:

- L_0 **norm** counts the number of features modified. For an image with pixel values x and adversarial image x', $\|x - x'\|_0$ equals the count of pixels that differ between the two images. An L_0 attack might change only a small number of pixels but by arbitrary amounts. This metric is relevant when adversaries can precisely manipulate specific features (e.g., changing a few critical pixels in an image or specific words in text).
- L_2 **norm** (Euclidean distance) measures the root-mean-square deviation: $\|x-x'\|_2 = \sqrt{\sum_i (x_i - x'_i)^2}$. This metric treats perturbations as vectors in high-dimensional space and measures their geometric length. L_2 perturbations tend to apply small changes across many features, producing smoother, more diffuse modifications that may appear as slight blurring or color shifts in images.
- L_∞ **norm** (maximum norm or Chebyshev distance) measures the largest single perturbation across all features: $\|x - x'\|_\infty = \max_i |x_i - x'_i|$. This bounds the maximum change to any individual feature, ensuring that no single pixel (in images) or feature (in other domains) changes by more

than a specified threshold. L_∞ constraints are commonly used because they provide intuitive bounds on per-pixel modifications.

Understanding these metrics is essential because different attack methods optimize for different norms, and defenses effective against one type of perturbation may fail against others.

Moosavi-Dezfooli et al. [5] introduced DeepFool, which frames adversarial example generation as finding the minimal perturbation required to cross the model's **decision boundary**, which is the geometric surface in input space that separates different classification regions. A classifier assigns class labels by determining which side of this boundary an input falls on; DeepFool iteratively computes the shortest path to cross it. By linearizing the decision boundary locally and solving for the minimal perturbation, DeepFool produces adversarial examples with smaller L_2 norms than FGSM, making them less perceptible while remaining effective.

Carlini and Wagner [6] developed a family of optimization-based attacks (known as C&W attacks) that reformulate adversarial example generation as a constrained optimization problem. Their approach jointly optimizes two objectives: (1) minimizing the distance between the original and adversarial example according to a chosen L_p norm, and (2) maximizing the confidence of misclassification. By carefully designing the optimization objective and using iterative gradient-based solvers, C&W attacks achieve state-of-the-art attack success rates across L_0, L_2, and L_∞ distance metrics. Notably, C&W attacks successfully defeated several defenses that had previously resisted simpler attacks, including **defensive distillation** [7], a technique that trains models on softened probability outputs rather than hard labels to smooth the loss surface and make gradient-based attacks less effective.

2.2.2 The Breakdown of Early Defenses

The proposal and subsequent failure of early defenses illustrated the challenge of defending against adaptive adversaries. Carlini and Wagner [8] conducted a systematic evaluation of ten detection-based defenses, which are methods designed to identify and reject adversarial inputs before they reach the classifier. They demonstrated that all ten defenses could be circumvented by adaptive attacks that account for the detection mechanism. This work established an important methodological principle: defenses must be evaluated against adaptive adversaries who know the defense strategy and can optimize attacks accordingly, not just against fixed attack methods developed before the defense was proposed.

This pattern repeated across numerous defense proposals. Defenses based on input transformations (such as JPEG compression, bit-depth reduction, or spatial smoothing) were bypassed by attacks optimized to remain effective after the transformation [9]. Defenses that obfuscated gradients by making it difficult for attackers to compute useful gradient information were shown to provide a false sense of security, as attackers could use gradient-free methods, approximate gradients, or exploit the defense mechanism itself [9].

2.2.3 Adversarial Training and Certified Defenses

As detection-based and gradient-masking defenses failed, the research community increasingly focused on two more principled approaches: adversarial training and certified robustness.

Adversarial training directly incorporates adversarial examples into the training process, teaching models to correctly classify both natural and adversarially perturbed inputs. Madry et al. [10] formalized

this approach as a **min-max optimization problem**:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim D} \left[\max_{\|\delta\|\leq\epsilon} \mathcal{L}(f_\theta(x+\delta), y) \right]$$

where θ represents model parameters, (x, y) are training examples from distribution D, δ is the adversarial perturbation bounded by radius ϵ, and $\mathcal{L}$ is the loss function. The inner maximization finds the worst-case perturbation within the allowed threat model, while the outer minimization trains the model to minimize loss even against these worst-case perturbations. This formulation provides a theoretically grounded framework for robustness, though adversarially trained models often exhibit reduced accuracy on clean (non-adversarial) inputs, a phenomenon known as the robustness-accuracy trade-off [11, 12].

Tramèr et al. [13] introduced **ensemble adversarial training**, which augments training data with adversarial examples generated from multiple source models (an ensemble) rather than only the model being trained. This approach improves robustness against black-box transfer attacks, where adversarial examples crafted for one model are used to attack a different model. By exposing the model to adversarial examples from diverse architectures during training, ensemble adversarial training reduces the model's susceptibility to transferred perturbations.

Certified robustness takes a fundamentally different approach by providing mathematical guarantees that a model's prediction cannot be changed by any perturbation within a specified radius, regardless of the attack method used [14]. Rather than empirically testing defenses against known attacks, certified defenses use formal verification techniques, randomized smoothing [15], or interval bound propagation [16] to prove that no adversarial example exists within a defined threat model. For instance, a certifiably robust classifier might guarantee that for a given input, any perturbation with L_2 norm less than 0.5 will not change the predicted class. While certified robustness offers strong security guarantees, these methods face scalability challenges: certification becomes computationally expensive for large models and high-dimensional inputs, and the radius of certifiable robustness is often smaller than what adversarial training can achieve empirically. The choice between empirical robustness (adversarial training) and certified robustness represents a trade-off between practical effectiveness and theoretical guarantees.

2.3 Expanding Threat Landscape

The field then expanded beyond evasion attacks to include privacy violations and threats during training. Membership inference attacks [17, 18] determine whether specific data points were part of model training. Carlini et al. [19] argued for assessing such attacks at low false positive rates, which reflect the serious nature of privacy issues.

Training-time attacks interfere with the learning process itself. Backdoor attacks introduce hidden triggers that cause specific misbehaviors when activated, with recent studies showing attacks on self-supervised learning [20], along with distribution-preserving methods that avoid detection [21]. Poisoning attacks that corrupt training data demonstrate that even small amounts of malicious samples can significantly harm model performance [22].

Physical-world attacks took gradient-based techniques beyond digital settings. Kurakin et al. [23] showed that adversarial examples remained effective when printed and photographed. Later work [24] developed robust physical adversarial examples against traffic sign recognition systems in autonomous

vehicles, raising questions about whether real-world attackers would use such complex methods when simpler options might work.

2.4 The Theory-Practice Gap

As the theoretical level of adversarial ML research increased, worries grew about whether this work addressed actual security needs in the real world. Kumar et al. [25] interviewed practitioners from 28 organizations, discovering that the industry lacked practical tools for dealing with adversarial ML threats. Grosse et al. [26] surveyed 271 industrial practitioners, revealing that while academic threat models were theoretically valid, research often overestimated what attackers could do, especially regarding access to training data and query limits.

Mink et al. [27] identified organizational barriers to deploying defenses through interviews with ML practitioners. These barriers included a lack of motivation from institutions, difficulty assessing AML risk, and competing business priorities. Practitioners often saw security as outside their expertise, revealing mismatches between job definitions in ML and security duties.

Apruzzese et al. [1] crystallized these issues through real-world case studies showing that actual ML system breaches often involved simple tactics rather than complex gradient-based attacks: input manipulation without gradients, exploiting system weaknesses instead of model-specific faults, operational gaps, and social engineering. This historical trend encourages a systematic analysis of recent publications to see if current research is addressing the theory-practice gap or continuing past patterns.

2.5 Types of Adversarial Attacks

Understanding the landscape of adversarial threats requires a taxonomy of attack objectives and methodologies. The adversarial ML community has identified distinct categories of attacks based on when they occur in the machine learning pipeline and what adversarial goals they serve [28]. These categories differ fundamentally in their threat models, required adversarial capabilities, and real-world implications for deployed systems.

Adversarial attacks generally fall into three primary categories based on their goals and the phase of the ML pipeline they target:

Evasion attacks manipulate inputs during inference to cause misclassification [2, 3]. The attacker changes an input (such as an image, malware sample, or network packet) so that the model gives an incorrect prediction while the model itself remains unchanged. These attacks target the inference phase and represent the most extensively studied threat type in adversarial ML literature. Early demonstrations by Szegedy et al. [2] and subsequent refinements [5, 6] established evasion attacks as the canonical adversarial threat, though their prevalence in academic research may not reflect their occurrence in real-world deployments.

Poisoning attacks corrupt the training process by injecting malicious data [29]. By adding carefully crafted malicious examples to training data, attackers can lead models to learn incorrect behaviors or establish backdoors, which are hidden triggers that cause specific failures when activated [30]. A poisoned model might classify most inputs correctly but systematically fail on inputs containing a particular pattern or trigger. Unlike evasion attacks that affect individual predictions, poisoning attacks compromise the model itself, potentially affecting all future inferences. Recent work has demonstrated poisoning

attacks against federated learning [31], self-supervised learning [20], and even distribution-preserving poisoning that evades detection [21].

Privacy attacks extract sensitive information from trained models [17]. Membership inference attacks determine if specific individuals were included in the training data [17, 18], potentially violating privacy even when models do not explicitly memorize training examples. Model extraction attacks aim to replicate model functionality through repeated queries [32], threatening intellectual property and enabling subsequent attacks. Data reconstruction attacks attempt to retrieve actual training examples from model parameters [33], with recent work demonstrating extraction of training data from large language models. These attacks raise fundamental questions about the privacy guarantees of machine learning systems and have motivated research into differential privacy [34] and other privacy-preserving training methods.

2.6 Threat Models and Their Implications

A critical dimension for evaluating adversarial ML research is the threat model, which formally specifies an adversary's capabilities, knowledge, and objectives [4]. The threat model determines what the attacker knows about the target system, what actions they can perform, and what constraints they face. This aspect represents a major area where academic research assumptions diverge from real-world deployment realities, as the adversarial capabilities assumed in research often exceed what attackers have in practice [1, 26].

The security community traditionally categorizes adversarial access along a spectrum from complete knowledge to query-only interaction:

White-box access assumes the attacker possesses complete knowledge of the model: its architecture, all parameters (weights and biases), training procedures, and possibly the training data itself [6, 10]. With white-box access, attackers can compute gradients (the mathematical derivatives indicating how changes to inputs affect model outputs), enabling efficient optimization-based attacks. Many seminal academic attacks assume white-box access [2, 3, 6] because it simplifies the mathematical formulation and enables rigorous worst-case analysis. However, this assumption provides adversaries with capabilities rarely available against production systems, where models are deployed as black-box services or protected by intellectual property measures.

Black-box access constrains the attacker to query-only interaction: they can submit inputs and observe outputs (predictions, confidence scores, or rankings) but cannot inspect internal model components [35]. This scenario more closely reflects real-world deployments where models operate as web services, mobile applications, or embedded systems with no internal visibility. Black-box attackers must rely on query-based optimization, transfer attacks from surrogate models [35], or input transformations that do not require gradient information [36]. Commercial ML services typically implement rate limiting, query costs, and anomaly detection that further constrain black-box attack feasibility.

Gray-box access represents intermediate scenarios where attackers possess partial knowledge [35]. They might know the model architecture but not the trained parameters, have access to a similar (but not identical) training dataset, or understand the training procedure without accessing the final model. Gray-box settings attempt to bridge the gap between white-box theoretical analysis and black-box deployment realities, though they remain less studied than the two extremes.

The choice of threat model profoundly affects research conclusions. Apruzzese et al. [1] stress that

real-world attackers rarely have white-box access to production systems, which typically employ API abstractions, model encryption, and restricted query interfaces. Similarly, Grosse et al. [26] document through practitioner surveys that academic assumptions about attacker knowledge, particularly regarding training data access and unlimited queries, systematically overestimate real-world adversarial capabilities. Yet, as our analysis shows, 63.2% of papers in our dataset assume white-box access, suggesting that research priorities may emphasize theoretical worst-case analysis over empirically grounded threat modeling.

3 Methodology

3.1 Data Collection

We systematically reviewed all papers with adversarial ML focus published at four top-tier security venues from 2022 through 2025. These venues were selected because they represent primary publication outlets for security-focused ML research and have historically shaped the field's direction.

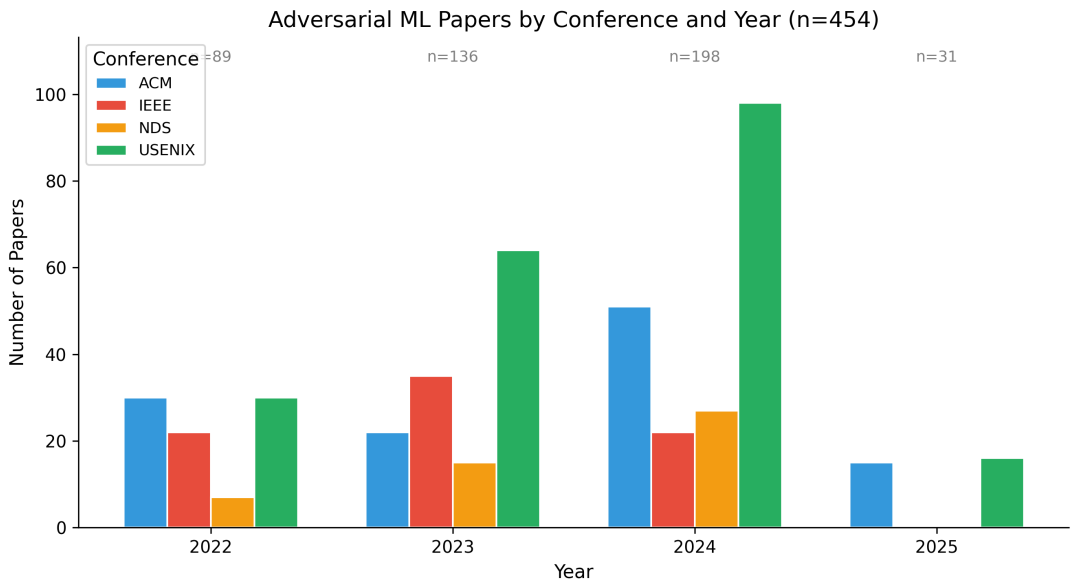

Figure 1: Dataset overview showing the distribution of 454 papers across four security conferences (ACM CCS, IEEE S&P, NDSS, USENIX Security) from 2022 to 2025.

Our dataset comprises 454 papers: ACM CCS contributed 118 papers (26.0%), IEEE S&P contributed 79 papers (17.4%), NDSS contributed 49 papers (10.8%), and USENIX Security contributed 208 papers (45.8%). The temporal distribution includes 89 papers from 2022, 136 from 2023, 198 from 2024, and 31 from 2025 (partial year at time of analysis).

3.2 Coding Framework

Each paper was evaluated across multiple dimensions designed to capture practical relevance:

- **Research Focus (G1):** Whether the paper primarily proposes attacks (60%), defenses (39%), or both (2%).

- **Attack Type (G2):** Classification as evasion (48%), poisoning (20%), privacy (23%), or multiple types (9%).
- **Data Domain (G4):** The input modality: images (65%), text (11%), audio (7%), malware (6%), or other (12%).
- **Threat Model (T1):** Assumed adversary access: white-box (63.2%), black-box (34.1%), or gray-box (2.6%).
- **Gradient Requirements (Q1):** Whether the approach requires gradient access.
- **Query Budget (Q2):** High (>1000 queries), low, or none.
- **Real System Testing (G7):** Whether validation occurred on deployed systems.
- **Code Release (G6):** Whether code was released publicly.

3.3 Gap Score Framework

To quantify the theory-practice gap, we developed a 6-point "Gap Score" summing binary indicators of assumptions that may limit practical applicability:

1. Requires white-box access (vs. black/gray-box)
2. Requires gradient computation
3. Assumes high query budget (>1000 queries)
4. Requires substantial computation (GPU-level resources)
5. No testing on deployed systems
6. No consideration of economic factors

Higher scores indicate greater distance from practical deployment considerations. A paper scoring 0 would employ realistic threat models, require minimal resources, validate on production systems, and consider economic constraints. A paper scoring 6 would represent a primarily theoretical contribution with limited immediate applicability to deployed systems.

4 Quantitative Findings

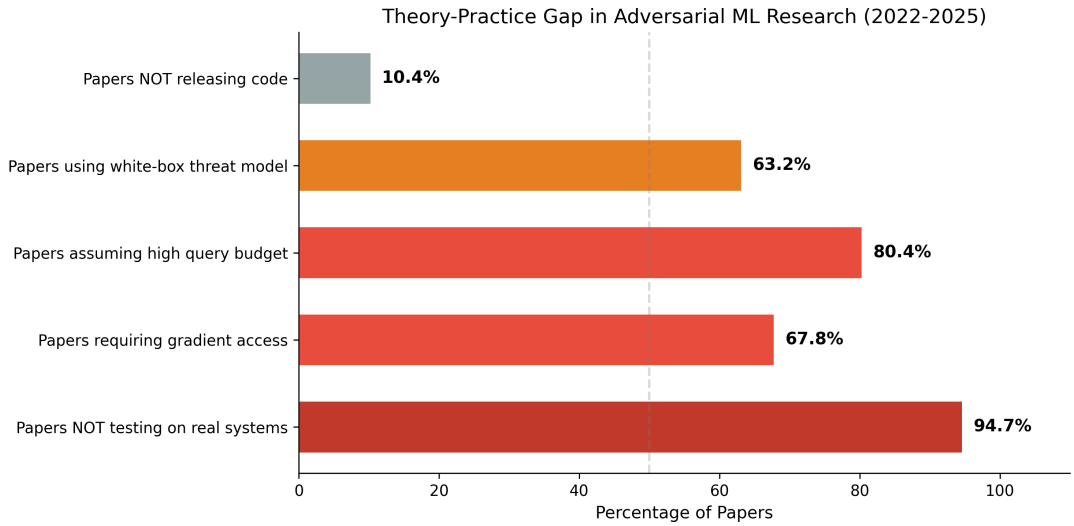

Figure 2: Overview of the theory-practice gap showing the percentage of papers exhibiting each characteristic: no real-world testing (94.7%), high query budget (80.4%), gradient dependency (67.8%), white-box access (63.2%), and no code release (10.4%).

Our analysis identifies several areas where research practices diverge from deployment realities. Figure 2 summarizes the key indicators across all 454 papers.

4.1 Real-World Validation

The most pronounced finding concerns real-world validation: **only 5.3% of papers (24 of 454) evaluate on actual deployed systems**. The remaining 94.7% validate exclusively on research benchmarks, simulated environments, or research prototypes.

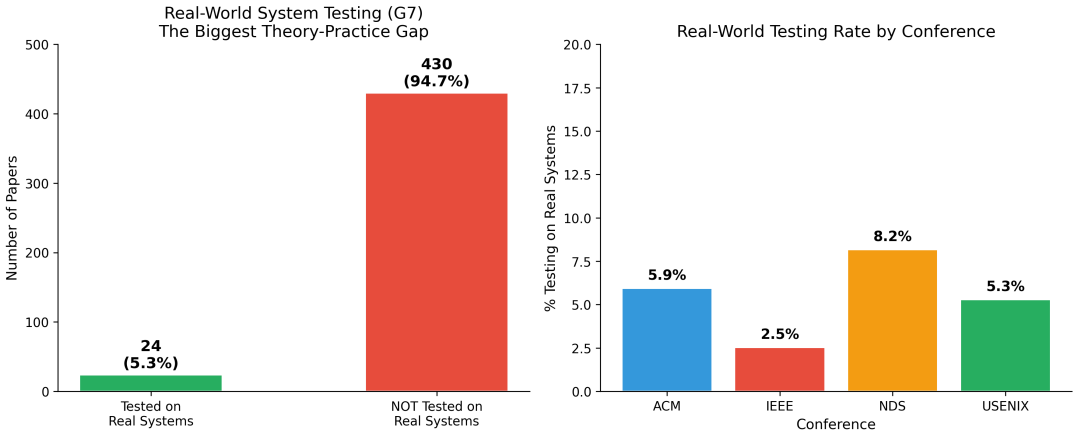

Figure 3: Real-world testing rate: 5.3% of papers validate on deployed systems.

This pattern has implications because deployment introduces constraints absent from controlled experiments. Production systems employ proprietary model formats and encryption. They integrate ML components into complex security pipelines where multiple components interact. They face hardware

constraints, latency requirements, and regulatory compliance obligations. Research validated only in laboratory settings may require substantial adaptation for deployment.

The few papers that do validate on real systems often reveal differences between laboratory and field performance. Nayan et al. [37] conducted a systematic review of on-device ML model extraction attacks, finding that many proposed academic attacks proved difficult to reproduce, performed less effectively on production models, or introduced unacceptable computational and energy costs. Similarly, Layton et al. [38] demonstrated that deepfake detection research employs metrics and dataset distributions that may lead to overestimation of detector efficacy.

Duan et al. [39] validated perception-aware attacks against YouTube's copyright detection system, representing one of the rare examples of testing against commercial infrastructure. Their work demonstrated that attacks optimized in simulation required adaptation to succeed in practice.

4.2 Gradient Dependency

Apruzzese et al.'s critique centered on the observation that "real attackers don't compute gradients." Our analysis indicates this pattern persists: **67.8% of papers require gradient access**, despite this capability being unavailable against most production systems.

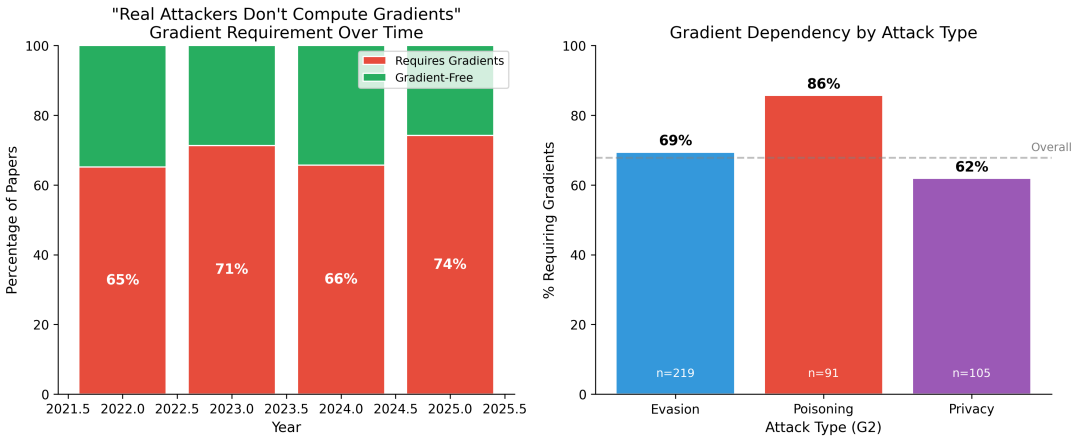

Figure 4: Gradient dependency analysis showing limited change over time (68.2% in 2022 to 65.9% in 2025) and variation by attack type.

Temporal analysis reveals limited change in this dimension. Gradient dependency has remained relatively stable: 68.2% in 2022, 69.1% in 2023, 66.8% in 2024, and 65.9% in 2025. While individual papers have explored gradient-free approaches, the overall distribution has not shifted substantially.

Some research has pioneered gradient-free methods in specific domains. The Universal Robustness Evaluation Toolkit (URET) from Eykholt et al. [36] formulates adversarial generation as a graph exploration problem, seeking sequences of domain-specific, functionality-preserving transformations rather than relying on differentiable feature spaces. This framework enables evaluation of systems processing inputs like malware binaries or tabular data where semantic and functional correctness must be maintained during perturbation.

Similarly, practical LLM jailbreak attacks demonstrate that effective attacks need not be gradient-based. Liu et al. [40] and Yu et al. [41] showed that jailbreaking large language models can be effective

even when executed via strategically crafted natural language prompts, illustrating the potential of accessible black-box attacks compared to complex gradient optimization.

4.3 Threat Model Assumptions

White-box access remains the predominant assumption: **63.2% of papers assume adversaries have complete knowledge of model architecture and parameters**. Only 34.1% consider black-box scenarios more closely matching deployment conditions, and 2.6% examine gray-box settings.

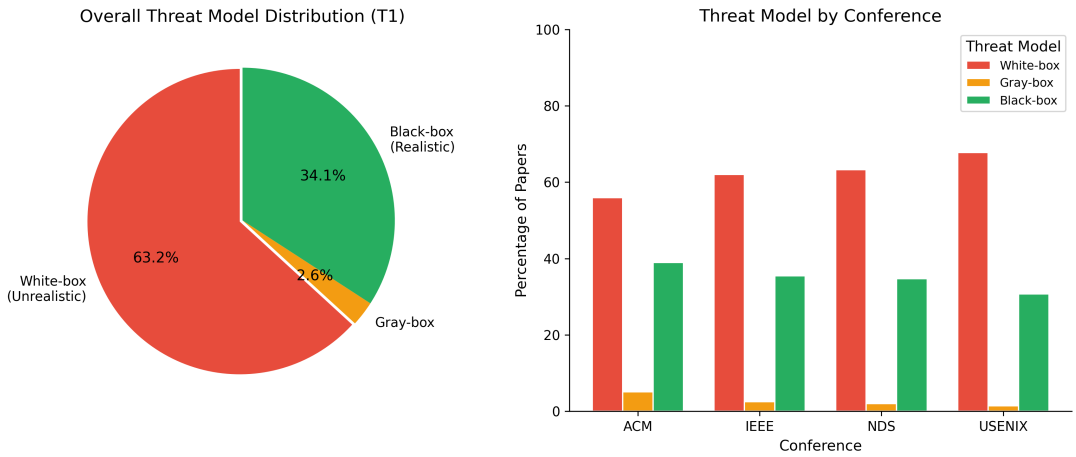

Figure 5: Distribution of threat model assumptions across all papers, showing prevalence of white-box assumptions.

This pattern contrasts with industrial practice. As Grosse et al. [26] document, academic studies often operate under assumptions of attacker access (such as extensive access to internal models, parameters, or training data) that do not reflect the security controls present in production environments.

The foundational meta-analysis by Arp et al. [42] identified methodological patterns in security research, including reliance on laboratory-only evaluation and deployment of threat models that may not account for adaptive adversaries. Our analysis suggests these patterns persist in the 2022–2025 literature.

4.4 Query Budget Assumptions

Even papers employing black-box threat models often assume substantial query access. **80.4% of papers assume high query budgets (>1000 queries)**, which may be impractical when commercial APIs implement rate limiting, repeated queries trigger anomaly detection, and real-time constraints limit iterative optimization.

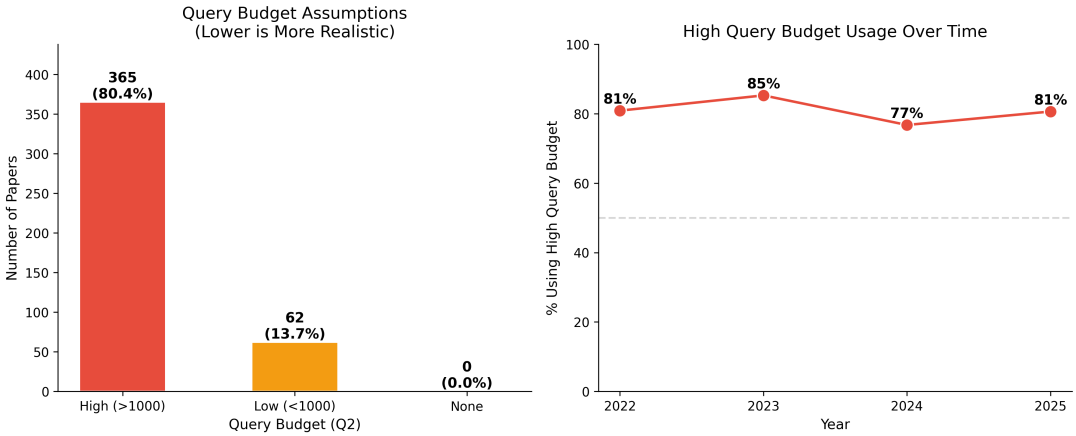

Figure 6: Query budget assumptions showing 80.4% of papers assume high query budgets.

Some recent work has addressed query efficiency. HARDBEAT from Tao et al. [43] generates triggers requiring knowledge only of the final predicted label (hard-label) and minimal queries, addressing restrictions imposed by commercial services. BounceAttack from Wan et al. [44] demonstrates query-efficient decision-based attacks. However, these remain exceptions rather than the predominant approach.

4.5 Domain Distribution

Adversarial ML research exhibits notable domain concentration: **65% of papers focus exclusively on image data**. Text receives 11% of attention, audio 7%, malware 6%, with all other domains combined representing 12%.

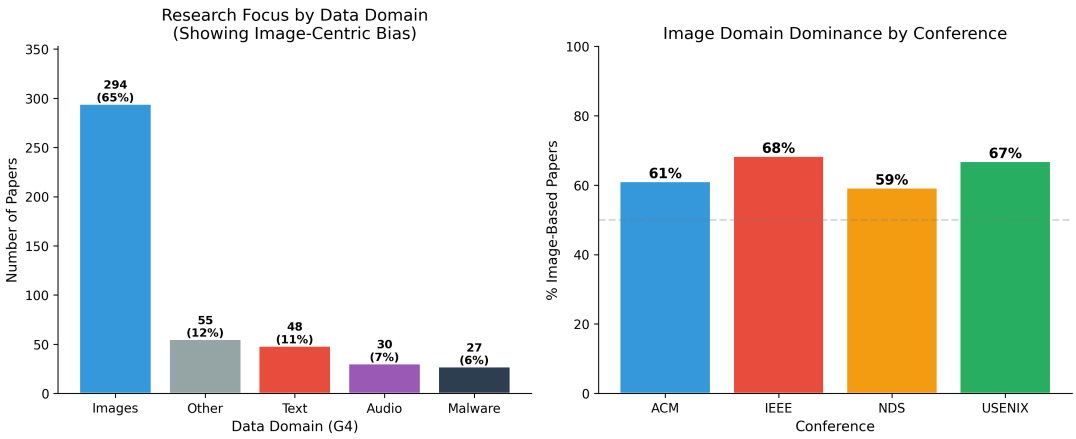

Figure 7: Data domain distribution revealing concentration on image classification in adversarial ML research.

This concentration creates potential blind spots. Financial systems process tabular data where adversarial perturbations cannot be measured by pixel distances. As Kireev et al. [45] observe for fraud detection, the meaningful constraint is not visual imperceptibility but rather the quantifiable financial cost or utility an adversary must expend. L_p norms may be less informative for tabular financial data; what matters is whether fraudulent transactions remain economically viable.

4.6 Code Availability

One dimension shows encouraging results: **89.6% of papers release their code**. This represents substantial commitment to reproducibility and exceeds code release rates in many other fields.

However, this introduces a nuanced consideration: code designed for research datasets may not transfer directly to production environments without substantial re-engineering. High code release rates support reproducibility while not necessarily indicating deployment readiness.

4.7 Gap Score Distribution

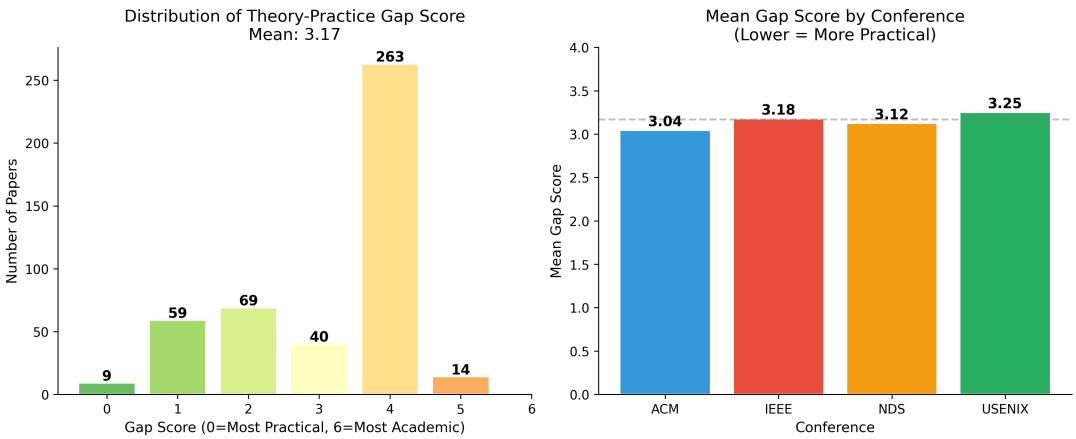

Figure 8: Distribution of Gap Scores showing the typical paper (mean 3.17/6) incorporates roughly half of the measured assumptions.

The mean Gap Score across all 454 papers is **3.17 out of 6**, indicating the typical paper incorporates approximately half of the assumptions we measured. The distribution peaks at scores of 3 to 4, with approximately 10.5% of papers achieving scores of 0 to 1 that would indicate closer alignment with deployment considerations.

Conference-level analysis reveals similar patterns across venues: ACM CCS averages 3.04, NDSS 3.12, IEEE S&P 3.18, and USENIX 3.25. No venue has established itself as substantially more practice-focused than others; the patterns we observe appear consistent across the security research community.

5 Thematic Findings

Beyond quantitative metrics, thematic analysis across venues reveals structural patterns in how adversarial ML research is conducted and evaluated.

5.1 The Utility-Robustness Trade-off

A significant consideration for deployment is the relationship between security guarantees and system performance. Research from USENIX Security (2022–2025) particularly illuminates this tension.

Xiang et al. [46] found that defensive proposals achieving certifiable robustness against adversarial patches frequently yielded reduced clean classification accuracy, which may discourage real-world deployment. By 2024, Xiang et al. [47] documented that certifiably robust defenses in computer vision

require 10 to 100 times more inference-time computation than undefended models, presenting computational challenges for practical deployment.

This pattern extends beyond vision systems. Ahmed et al. [48] found that defenses against malicious activations in voice assistants typically affect natural accuracy. Similarly, widely used privacy-preserving techniques like DP-SGD may compromise model utility to achieve privacy guarantees [49, 50].

Some recent work demonstrates progress in addressing this trade-off. PatchCleanser introduced a double-masking approach compatible with any image classifier, achieving high certified robustness while preserving state-of-the-art clean accuracy [46]. MIST offers a pathway toward robust security by strategically limiting overfitting only to the most membership-vulnerable training instances [51]. CAMP training demonstrated that provable adversarial robustness in deep reinforcement learning need not sacrifice certified expected return, which may be valuable for safety-critical robotics applications [52].

5.2 Evaluation Methodology

Research across venues reveals considerations regarding how attacks and defenses are evaluated.

5.2.1 Distance Metrics

Standard evaluation measures adversarial perturbation size using L_p norms, which are mathematical measures of distance between original and perturbed inputs. However, as Carlini et al. [19] document, conventional distance metrics like L_p norms, which measure pixel-level differences, do not reliably predict whether humans perceive adversarial perturbations.

This consideration was empirically examined by the Avara framework [53], which used VR environments and eye-tracking to study whether drivers notice adversarial traffic signs. They found that L_p norms do not reliably predict whether human drivers notice adversarial perturbations: attacks deemed "imperceptible" by mathematical standards may be immediately obvious to a driver, while attacks violating L_p constraints might go unnoticed in realistic driving conditions.

5.2.2 Privacy Evaluation

Privacy attacks present particular evaluation challenges. Carlini et al. [19] observe that privacy is fundamentally a worst-case concern: a defense succeeds only if it protects all individuals, not just the majority. Yet membership inference attacks are typically evaluated using average-case metrics (overall accuracy, AUC) that may mask severe privacy leakage for specific individuals.

5.3 Physical Deployment Considerations

A consideration for conventional AML research concerns the gap between digital adversarial examples and the complex physical conditions governing real-world perception systems.

5.3.1 Physical Attack Challenges

Digital perturbations optimized in simulation may perform differently when deployed physically due to environmental factors: distance and viewing angle variations, illumination changes, sensor noise, and

compression artifacts. NDSS research has particularly emphasized this consideration.

Jia et al. [24] developed robust physical adversarial example pipelines tested against production autonomous vehicles running YOLO v5 traffic sign recognition. Their work required accounting for real-road conditions that simulation may not capture.

Physical attack research at USENIX has expanded beyond vision systems. Liu et al. [54] designed physically realizable 3D adversarial objects capable of deceiving X-ray prohibited item detection, requiring optimization for shape rather than color or texture and accounting for complex object overlap in luggage. Cao et al. [55] demonstrated Physical Removal Attacks using focused laser spoofing to selectively remove LiDAR point cloud data on autonomous vehicles. The "Tubes Among Us" research [56] demonstrated analog adversarial attacks where adversaries manipulate voice signals using simple tubes to bypass speaker recognition, effectively bypassing established digital artifact detection methods.

By 2025, physical attack research has embraced increasingly practical scenarios. "Shadow Hack" [57] exploits LiDAR weaknesses using ordinary non-reflective materials placed on roads, requiring no specialized equipment. ATKSCOPES [58] demonstrates rapid evasion against real-world perceptual hashing algorithms by dynamically adapting to victim systems.

5.3.2 System Integration

Research increasingly recognizes that integrating ML models into complex systems introduces considerations that isolated model analysis may miss. Debenedetti et al. [59] reveal that system-level components such as training data filters or output monitoring may introduce privacy side channels exploitable by adaptive adversaries, potentially affecting provable differential privacy guarantees.

Nasr et al. [60] demonstrated this by examining how exploiting Google's Magika file-type classifier affects Gmail's malware detection pipeline, showing that a single non-robust component can compromise an entire security pipeline.

Chen et al. [61] document that security mechanisms integrated solely at the framework level may not persist once models are compiled into optimized executables for deployment. Defenses may need to be embedded within the DL compiler pipeline to persist through deployment, a requirement often absent from academic work.

5.3.3 Real-World Adversarial ML Scenarios

Recent incidents reveal how adversaries exploit the interfaces and economics of deployed ML systems rather than computing gradients. The 2025 *EchoLeak* attack showed how a single phishing email with hidden instructions could trigger a chain reaction: the attacker embedded subtle prompts instructing Microsoft 365 Copilot not to mention the email's content, inserted a reference-style Markdown link that bypassed link filtering, and relied on the Teams client's automatic image preview to fetch data via an allowed proxy, resulting in zero-click exfiltration [**?**]. A similar prompt-injection exploit on Google Bard Extensions placed malicious instructions in a shared document; when the model summarised the file, it encoded private data into an image URL and used Google Apps Script to evade content-security policies [**?**]. Researchers further demonstrated with *AgentFlayer* that a single poisoned document uploaded to a Google Drive could extract API keys when ChatGPT's Connector summarised it [**?**], and Tenable's "HackedGPT" study exposed how ChatGPT's browsing and memory features can be hijacked through comments on trusted websites or cleverly crafted URLs. Collectively, these cases

underscore that LLM deployments are vulnerable to prompt injection through email clients, document summarisation and web-browsing contexts, and that attackers chain low-cost techniques across multiple system components.

Beyond prompt injection, adversaries have shown that they can extract models and data through repetitive queries. The "divergence attack" on ChatGPT causes the aligned `gpt-3.5-turbo` model to behave like its unaligned base: by prompting the system to repeat a word indefinitely, researchers coaxed it into revealing training data (including real email addresses and phone numbers) at a rate 150 times higher than previous attacks [**?**], recovering thousands of training examples for about $200 in query costs. Other studies of model extraction against ML-as-a-service APIs report fidelities between 63 % and 84 % when cloning commercial emotion-recognition models [**?**], indicating that modest query budgets suffice to approximate proprietary systems. These findings reveal that alignment and rate-limiting measures alone do not prevent data leakage or model theft.

Finally, adversaries are attacking the ML supply chain and human layer. The *Nightshade* tool allows

5.4 Attack Evolution

Despite methodological considerations about assumptions, research has progressively explored more practical attack vectors.

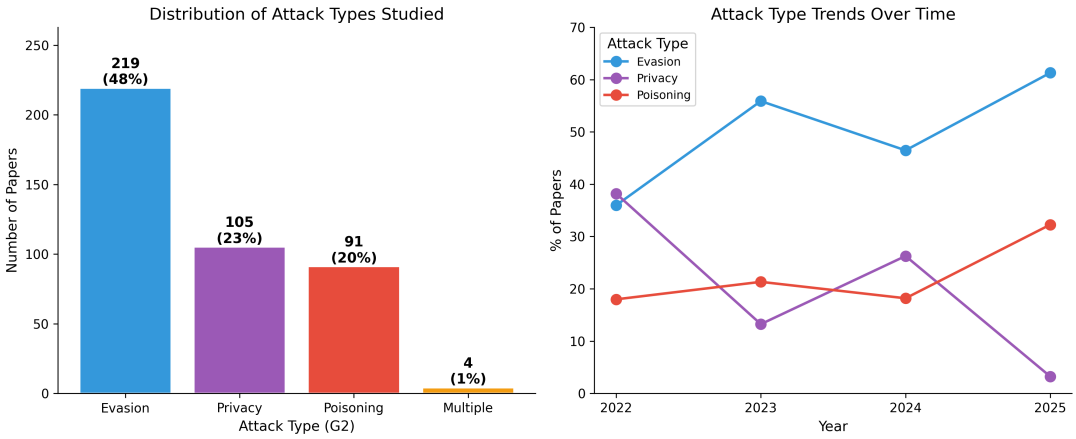

Figure 9: Attack type distribution showing evasion attacks (48%) predominate, with trends over time.

Early practical attacks (2022) demonstrated physically realizable approaches like the "frustum attack," which leverages environmental context to compromise automotive sensor fusion in black-box settings [62].

By 2023–2024, attacks increasingly emphasized simplicity and accessibility. The effectiveness of jailbreaking LLMs through natural language prompts, even by users without ML expertise, illustrated the potential of accessible black-box attacks compared to complex gradient optimization.

The 2025 landscape shows attacks prioritizing stealth and persistence. MergeBackdoor [63] reveals supply chain considerations where seemingly benign upstream models pass security checks but activate malicious backdoors upon merging with other components. Guo et al. [64] describe persistent backdoor strategies targeting stable neuronal components, ensuring exploits survive continuous parameter updates in continual learning systems.

5.5 Defense Evolution

Defensive research has evolved from generic solutions toward specialized, interpretable, and context-aware mechanisms.

Blacklight [65] detected and mitigated black-box query-based attacks against MLaaS by leveraging the observation that iterative optimization produces highly similar queries, providing defense against persistent attackers that bypass account-based security measures.

Recent defenses demonstrate increased sophistication. JBShield [66] moves beyond heuristics for LLM protection by using the Linear Representation Hypothesis to identify and manipulate "toxic" and "jailbreak" concepts within hidden states. SafeSpeech [67] proactively affects voice data during training to make synthesized audio less usable, achieving robustness against voice cloning. DeBackdoor [68] addresses deployment constraints by providing backdoor detection effective under black-box access, data scarcity, and pre-deployment inspection limitations.

5.6 Economic and Organizational Factors

Academic research has given limited attention to the economic and organizational dimensions of adversarial ML.

5.6.1 Economic Considerations

ACM CCS research has highlighted how commercial ML services create incentives for IP theft that may affect protective mechanisms. Cong et al. [69] document that extracting pre-trained encoders may cost substantially less than training from scratch. Lu et al. [70] show that Neural Dehydration can remove watermarks using less than 2% of training data. The economic asymmetry, where model extraction attacks may cost orders of magnitude less than defenses or the assets they protect, remains an area requiring additional attention.

5.6.2 Organizational Considerations

Beyond technical considerations, qualitative research reveals organizational factors affecting industry adoption. Mink et al. [27] found that ML practitioners often lack institutional motivation and resources to understand and mitigate adversarial threats, frequently viewing security and machine learning as disconnected fields.

This organizational separation results in lower prioritization of adversarial evaluations before deployment and limited visibility into monitoring for active attacks. Defenses may remain unimplemented due to isolation between ML and security teams or because competing business priorities outweigh the cost and time needed for robust implementation.

6 Foundation Models and Large Language Models: An Emerging Parallel Frontier

The rapid growth of large language models represents a significant shift in adversarial machine learning, developing alongside traditional research focused on computer vision. While image classifiers operate

in continuous pixel space with bounded L_p-norm perturbations, LLMs function in discrete token spaces with open-ended context windows and fundamentally different attack surfaces. These systems now power applications affecting millions, including customer service chatbots, code generation tools, and medical assistants. However, adversarial research in this domain reveals the same theory-practice gaps observed in traditional ML fields.

6.1 From Manual Jailbreaks to Automated Optimization

Initial LLM adversarial research relied on manual prompt engineering, where simple role-playing scenarios or encoded instructions (ROT13, Base64) bypassed safety measures. This changed with Zou et al.'s [71] Greedy Coordinate Gradient (GCG) attack, which automates discrete optimization over token sequences to generate universal adversarial suffixes. GCG achieved 100% success on Vicuna-7B and 88% on Llama-2-7B-Chat, with transfer rates of 87.9% for GPT-3.5, 53.6% for GPT-4, and 66% for PaLM-2. Notably, Claude-2 showed significantly higher resistance at only 2.1% success.

Yet GCG exemplifies the theory-practice gap. The attack requires white-box gradient access through the model's embedding layer and hundreds of thousands of forward and backward passes during optimization. For production APIs that expose only final text outputs, obtaining gradients is impossible. Carlini et al. [72] note that conducting such attacks would cost approximately $18 on GPT-3.5-Turbo, making them impractical when loss and gradients cannot be easily accessed.

Recent research has shifted toward black-box methods reflecting real-world constraints. Chao et al. [73] introduced PAIR (Prompt Automatic Iterative Refinement), using an attacker LLM to refine jailbreak prompts based on target model feedback, succeeding in fewer than 20 queries. Mehrotra et al. [74] developed Tree of Attacks with Pruning (TAP), achieving over 80% success against GPT-4-Turbo and GPT-4o while bypassing LlamaGuard. These represent serious attack vectors requiring only API access and operating within reasonable query limits.

Multi-turn attacks reveal particularly concerning vulnerabilities. Russinovich et al. [75] demonstrated the Crescendo attack, gradually escalating through seemingly harmless conversations to achieve up to 100% success in under 10 turns. Each message passes safety checks individually; only the accumulated context produces policy violations. Production chat interfaces designed to maintain conversation history inadvertently enable attacks that unfold over minutes rather than seconds.

6.2 Multimodal Models: Exploiting Cross-Modal Alignment Gaps

Vision-language models present attack surfaces absent in text-only systems. Qi et al. [76] showed that a single adversarial image can universally jailbreak vision-language models, forcing compliance with harmful instructions the models would reject in text-only contexts. This vulnerability arises from an alignment gap: while LLM components undergo extensive safety training on text data, vision encoders receive substantially less adversarial hardening.

Typography-based attacks achieve concerning results without gradient access. Gong et al. [77] introduced FigStep, converting harmful text into images that models read via OCR, yielding 82.5% average attack success across six open-source vision-language models compared to 44.8% for identical text-only queries. Against commercial systems, Niu et al. [78] demonstrated multi-modal linkage attacks combining encryption-decryption processes across modalities, achieving 97.8% success on GPT-4o's SafeBench, 98.8% on MM-SafetyBench, and 99.1% on HADES-Dataset.

Audio modalities show similar vulnerabilities. Zhang et al. [79] developed AdvWave, demonstrating that adversarial acoustic noise can jailbreak speech-enabled models with 40% higher success rates than baselines. It successfully jailbroke GPT-4o's speech-to-speech API in just 30 queries with nearly perfect results. The perturbations mimic common urban sounds (dog barks, car horns), making detection difficult without compromising legitimate functionality.

6.3 The Theory-Practice Gap Manifests Differently for LLMs

Examining LLM adversarial research through our analytical framework reveals familiar issues with domain-specific characteristics:

Model Access Assumptions. The most powerful attacks require white-box gradient access available only for open-source models. Production APIs like GPT-4 and Claude expose at most top-5 token probabilities, making direct loss calculations impossible. Huang et al. [80] found their "Catastrophic Jailbreak" achieved 95% misalignment on 11 open-source models but only 7% on ChatGPT, an 88-point gap demonstrating substantially more thorough safety training for proprietary models.

Query Budget Realism. OpenAI's free tier permits approximately 3 requests per minute, while paid tiers allow up to 3,500 requests per minute [81]. Gradient-based attacks requiring 50,000 to 256,000 queries face economic barriers (costs exceeding $18 per attempt) and rate limiting that academic papers often overlook. Black-box methods like PAIR operate within more realistic budgets (approximately 20 queries), though even these face practical limitations at scale.

Evaluation Instability. Unlike static computer vision models, LLM providers continuously update systems and silently patch vulnerabilities. A paper claiming 80% success against "GPT-4 as of March 2024" cannot be verified by June 2024 when the model has changed. Rando and Tramèr [82] note that this evolving defense landscape means "we cannot measure progress against continuously updated closed-source systems," creating benchmarking challenges absent in traditional adversarial ML.

Threat Model Formalization. LLM attacks lack the formal structure bounding CV adversarial examples (like L_p-norm budgets or imperceptibility constraints). Defining "harmful" output requires subjective human judgment. Perez et al. [83] showed that manual red teaming still outperforms automated methods in finding worst-case model failures, suggesting the full attack landscape remains only partially understood by algorithmic approaches.

Prompt-injection incidents further illustrate how the theory–practice gap for LLMs is rooted in interface design. EchoLeak bypassed Copilot's prompt classifier, link filter and content-security policy by chaining hidden instructions, reference-style links and auto-fetched images [**?**]. Bard's Extensions were hijacked by a hidden prompt in a shared document, causing the model to encode sensitive data into a URL [**?**]. *AgentFlayer* showed that a single malicious document could extract API keys via OpenAI's Connectors [**?**]. Tenable's HackedGPT study identified seven vulnerabilities in ChatGPT's browsing and memory tools, demonstrating how comments or trusted websites can inject instructions into the model's browsing context. These cases reveal that LLM robustness must address prompt filtering, memory management and connector policies alongside traditional model training.

6.4 Defense Mechanisms: Progress and Persistent Limitations

Production systems have adopted guardrail architectures assessing prompts and responses before and after LLM processing. Meta's Llama Guard [84] fine-tunes a Llama model to detect policy violations in

six harm categories, now at its fourth version with multimodal support. NVIDIA's NeMo Guardrails [85] provides programmable input/output guidelines using Colang, achieving 1.4× improvement in safety compliance with approximately 0.5 seconds added latency.

Recent studies reveal significant weaknesses. Hackett and Gretton [86] tested six guardrail systems, including Azure Prompt Shield and Meta Prompt Guard, against character injection attacks (homoglyphs, zero-width characters, Unicode tags) and emoji smuggling, finding up to 100% evasion success with no guardrail performing consistently across attack types. Perplexity-based detection works well against GCG-style attacks producing scrambled suffixes but fails against semantic jailbreaks generating fluent text.

Automated red teaming has advanced substantially. Bertrand et al. [87] introduced Rainbow Teaming, employing quality-diversity optimization through MAP-Elites to create diverse attack databases, achieving greater than 90% success across Llama 2/3 models while providing attacks valuable for diagnosis and safety fine-tuning. Hong et al. [88] applied curiosity-driven reinforcement learning to red teaming, successfully provoking harmful responses from RLHF-tuned models.

Bai et al. [89] introduced Constitutional AI, using RLAIF (RL from AI Feedback) with explicit constitutional principles for self-assessment and revision, producing responses that explain objections rather than refusing outright. However, conflicting goals between helpfulness and harmlessness create vulnerabilities that adversaries can exploit.

Standardized benchmarks aim to systematize jailbreak assessments. Chao et al. [90] released JailbreakBench, featuring 100 misuse behaviors across 10 OpenAI policy categories with public leaderboards tracking attack methods and defenses. While enabling reproducible comparison, it faces typical safety evaluation issues: subjective harm definitions, model updates invalidating findings, and challenges capturing worst-case behaviors through automated metrics.

Agent-specific vulnerabilities represent an emerging concern. Zhan et al. [91] developed InjecAgent, testing 1,054 cases for indirect prompt injection in tool-integrated agents, finding ReAct-prompted GPT-4 vulnerable 24% of the time, with enhanced attacks nearly doubling success rates. When LLMs control tools, access databases, and execute code, successful jailbreaks shift from producing problematic text to taking consequential real-world actions.

6.5 Foundation Model Properties Amplify Systemic Risk

Bommasani et al. [92] identified two features distinguishing foundation models from traditional ML: emergence (unpredictable capabilities from scale) and homogenization (single models serving numerous downstream applications). Adversarial weaknesses in GPT-4 impact not only ChatGPT but every application built on its API, affecting tools from legal document analyzers to educational tutors and medical triage systems. Unlike standalone image classifiers where a compromise affects specific implementations, vulnerabilities in foundation models have systemic effects across the ecosystem.

7 Temporal Trends

A central question motivates this review: has the research community responded to calls for greater practical relevance? Our temporal analysis provides insight into this question.

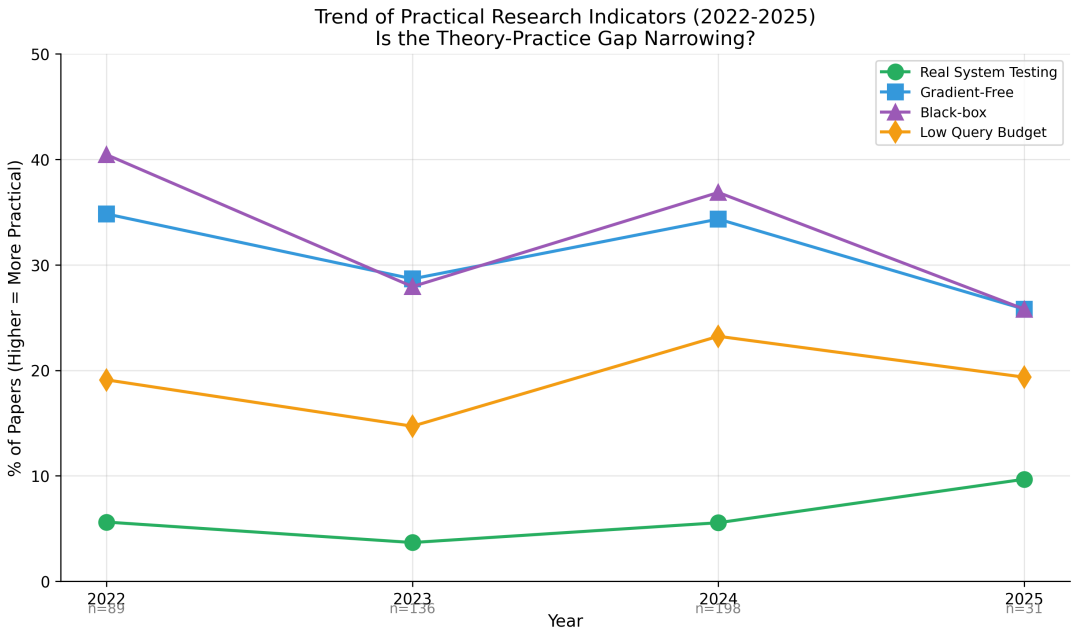

Figure 10: Temporal trends showing the theory-practice gap from 2022 to 2025, with limited change in key metrics.

7.1 Stable Patterns

Real-world testing remains at approximately 5% across all years. Despite calls for deployment validation, the research community has not substantially shifted toward testing on production systems.

Gradient dependency has remained relatively stable at 67–69%. There has been limited movement toward gradient-free approaches at the aggregate level.

White-box assumptions show no substantial reduction (63% in 2025 vs. 64% in 2022). Threat model distributions have remained consistent.

Domain distribution persists with image data representing 65% of papers throughout the study period.

Economic analysis remains below 11% in all years.

7.2 Areas of Modest Progress

Query efficiency shows some progress, with high-budget assumptions declining from 80.4% to approximately 76% by 2025. Some researchers have developed query-efficient attacks, though these represent a minority of approaches.

Awareness of attacker knowledge limitations has increased, with growing acknowledgment that adversaries often lack full system knowledge. However, this awareness has not yet translated into substantial shifts in threat modeling practices.

7.3 Emerging Research Areas

The 2024–2025 period has seen substantial growth in LLM security research, representing a significant shift in the adversarial ML landscape. While traditional research focused predominantly on computer

vision (65% of papers in our dataset), large language models and multimodal foundation models introduce fundamentally different attack surfaces. These systems operate in discrete token spaces rather than continuous pixel spaces, face evaluation instability as providers continuously update models, and lack the formal perturbation constraints (like L_p-norm budgets) that bound traditional adversarial examples.

Yet the theory-practice gap manifests similarly. White-box gradient-based attacks like GCG [71] achieve high success rates on open-source models but prove impractical against production APIs that expose only final outputs. Black-box methods like PAIR [73] and TAP [74] operate within more realistic constraints, succeeding in under 20 queries against commercial systems. Multimodal models reveal cross-modal alignment gaps, where vision encoders receive less adversarial hardening than text components [76]. As discussed in Section 6, these systems amplify systemic risk through homogenization: vulnerabilities in foundation models like GPT-4 affect every downstream application built on their APIs, from legal document analyzers to medical triage systems.

8 Cross-Venue Analysis

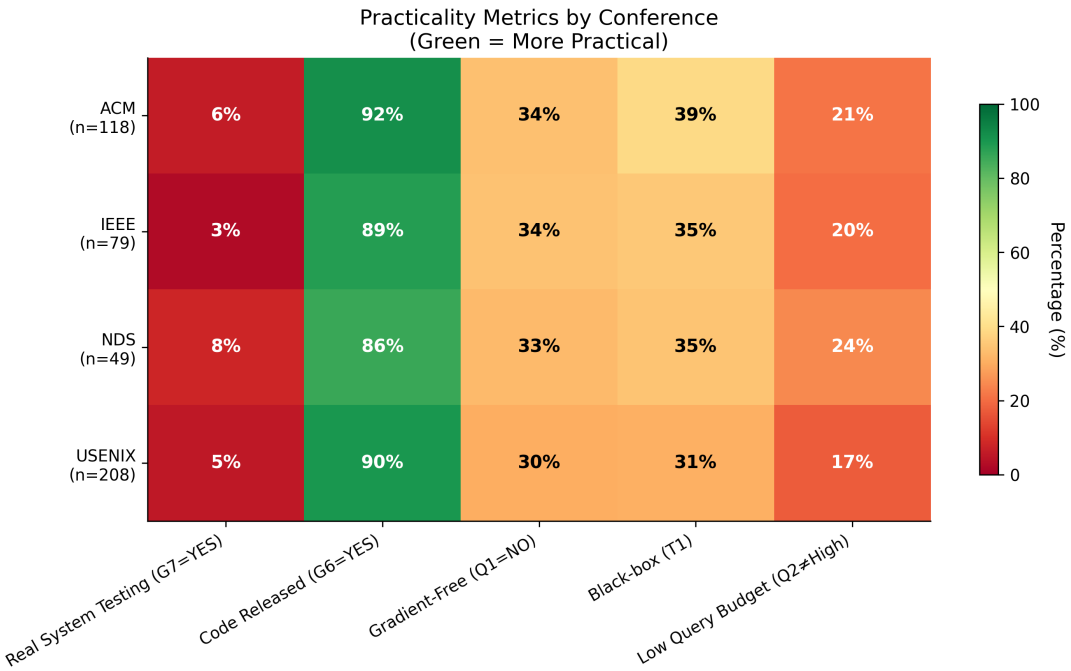

Figure 11: Conference comparison heatmap showing similar patterns across all venues.

While all venues share similar theory-practice gap patterns, each has developed distinctive emphases that collectively illuminate different facets of the research landscape.

8.1 ACM CCS: Economics and System Integration

ACM CCS research uniquely emphasizes business logic, usability constraints, and intellectual property protection. Contributions include demonstrating how economic incentives shape the threat landscape [69, 70], documenting relationships between mathematical metrics and human perception [53], analyzing system-level integration considerations [60], and examining architectural constraints in emerg-

ing deployment patterns.

8.2 IEEE S&P: Efficiency and Formal Guarantees

IEEE S&P research emphasizes computational efficiency, privacy-utility trade-offs, and certified scalability. The venue has advanced understanding of threat modeling considerations and resource assumptions [19], documented accuracy-privacy trade-offs [93], and examined scalability considerations for certified robustness deployment [94].

8.3 NDSS: Physical Constraints and Operational Realities

NDSS research emphasizes system-level constraints, distributed training considerations, and physical deployment realities. Contributions include documenting gaps between digital perturbations and physical deployment [24], developing metrics for non-image domains [45], analyzing distributed training heterogeneity [95], and studying adversarial dynamics where defenses create exploitable side channels.

8.4 USENIX Security: Real-World Validation and Domain Diversity

USENIX Security research emphasizes testing on deployed systems and domain-specific constraints. The venue has documented tensions between theoretical rigor and deployment constraints [46, 47], analyzed trade-offs involving computational cost, regulatory compliance, and usability [48], studied adversarial dynamics when defenses encounter adaptive attackers in production, and examined domain-specific requirements [36].

8.5 Attack versus Defense Papers

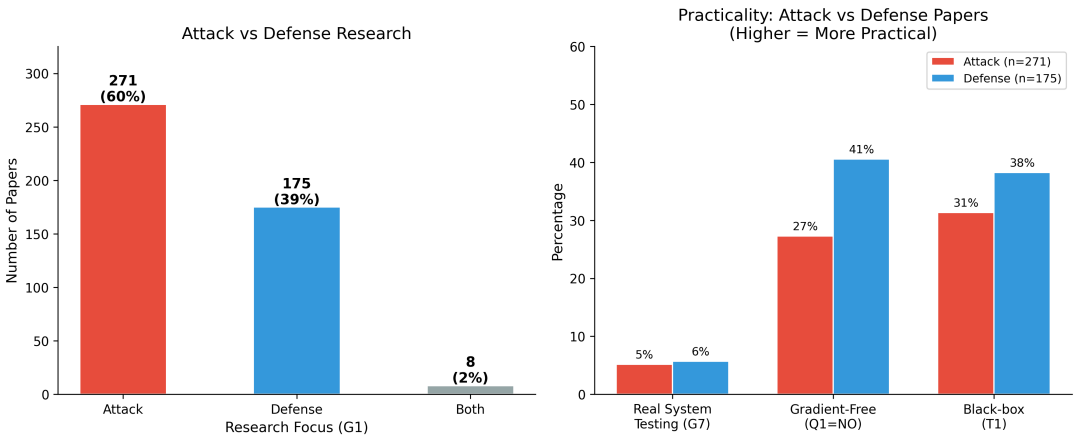

Figure 12: Comparison of practicality metrics between attack-focused and defense-focused papers.

Our analysis reveals that defense papers demonstrate somewhat higher alignment with practical considerations than attack papers on average. Defense papers show 6% real-system testing versus 5% for attacks, 41% gradient-free approaches versus 27%, and 38% black-box focus versus 31%. However, both categories remain distant from deployment-ready research.

9 Recommendations

Bridging the theory-practice gap requires coordinated effort across the research ecosystem. We offer recommendations for multiple stakeholder groups.

9.1 For Researchers

9.1.1 Methodological Considerations

Validate on real systems when feasible. Partner with industry to evaluate against production deployments. Even limited real-world testing can reveal constraints not apparent in laboratory settings.

Employ realistic threat models. Consider black-box access as a default assumption. When white-box assumptions are necessary, justify them explicitly and acknowledge their implications.

Expand domain coverage. Extend research beyond image classification to tabular data, graphs, time-series, multimodal systems, and large language models where practical deployment is occurring.

Consider economic factors. Analyze attack costs, defender resources, and return-on-investment trade-offs. Security decisions involve economic considerations.

Include human evaluation. Test perceptibility against actual humans rather than relying solely on mathematical metrics.

9.1.2 Evaluation Practices

Evaluate privacy using true-positive rate at low false-positive rates, not only average accuracy or AUC. For non-image domains, define domain-appropriate perturbation constraints. Report query budgets realistically based on what commercial APIs actually permit. Measure computational requirements in practical units such as wall-clock time and computational cost.

9.2 For Venues and Program Committees

Encourage real-world validation. At minimum, request authors to justify why real-system testing is infeasible if omitted.

Consider artifact badges for deployability. Recognize work tested on production systems with explicit acknowledgment.

Develop evaluation guidelines. Request papers to explicitly address threat model realism, query budget constraints, gradient requirements, domain-appropriate metrics, economic considerations, and adaptive adversary testing.

Value negative results. Papers demonstrating that attacks fail under realistic constraints provide valuable information that positive-result publication preferences may suppress.

9.3 For Industry Practitioners

Interpret academic results in deployment context. Adjust risk assessments for deployment realities rather than accepting laboratory success rates directly.

Anticipate adaptation requirements. Budget for re-engineering before research prototypes become production-ready.

Prioritize defenses against realistic threats: black-box attacks, low query budget scenarios, economically motivated adversaries, and domain-specific constraints.

Consider contributing data. Anonymized information about real attack attempts could help ground academic research in deployment reality.

9.4 For Funding Agencies

Support industry-academic partnerships with real-world validation components. Fund deployment studies rather than only paper publication. Incentivize replication studies that validate academic claims against production systems. Consider red team / blue team programs with realistic constraints.

10 Limitations and Threats to Validity

10.1 Scope Limitations

This review focuses on four security venues and may not capture patterns at ML conferences (NeurIPS, ICML, ICLR) where different norms may apply. Publication preferences may suppress negative results; attacks that fail under realistic constraints or defenses that prove impractical are less likely to be published. Our 2022–2025 window may not capture longer-term trends.

10.2 Coding Limitations

The Gap Score reduces complex trade-offs to binary decisions, potentially oversimplifying nuanced situations. Manual coding introduces subjectivity, particularly for papers spanning multiple categories. The definition of "real system testing" may vary; some papers test on emulated production environments that share some but not all deployment constraints.

10.3 Generalizability

Security venues may actually be more practice-focused than typical computer science venues given their traditional emphasis on real-world threats. Our findings may therefore underestimate the theory-practice gap in the broader ML research community.

11 Conclusion

This systematic review involving 454 research papers published from 2022–2025 at four prestigious security conferences bears evidence of some beneficial contributions by the research community of adversarial machine learning studies in spite of the theory-practice gap noticed by Apruzzese et al.

The quantitative trends are striking:

- 94.7% of the papers don't test on real systems
- 67.8% need gradients which might not be accessible in real life
- 80.4% assume query budgets that could exceed deployment limits
- 63.2% assume white-box access that could not be provided in deployment situations

Thematic analysis shows other considerations emerging. L_p norms may not model human perception well. Robustness may not generalize to production-quality settings. Defense systems do not necessarily take into consideration economic losses and use ease-of-use constraints. Assessment metrics are often averages for the worst-case scenario characteristics.

The emergence of large language models and foundation models represents a parallel frontier where these same theory-practice gaps manifest through different attack surfaces. While these systems introduce new challenges (discrete token spaces, evaluation instability, cross-modal vulnerabilities), they exhibit familiar patterns: impractical gradient-based attacks, unrealistic query budgets, and insufficient real-world validation. The systemic nature of foundation model deployment, where single models serve numerous downstream applications, amplifies the practical importance of addressing these gaps.

Among the many contributions of the research community is the theoretical foundation and pioneering research in adversarial phenomena. However, to move forward, accomplishing the goal of closing the theory-practice gap will necessitate structural changes in conferences that reward verification of effectiveness in the real world, assumptions of reasonable threats in research, industry inputs of deployment results, and distribution of funding that emphasizes effectiveness along with innovation.

However, the efficacy of ever more widespread ML-dependent systems is contingent on the ongoing involvements of the community of ML researchers with both the theory and the practicalities. Our observations indicate that those practicalities can, through shifts within publishing, evaluation, and collaboration, help ensure that community-driven ML adversarial research and development work increasingly serves a practical, security context.

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A Complete Paper Analysis Dataset

The complete analysis of all 454 papers is available in the supplementary CSV file: `all_conferences_analysis_r`

The dataset includes the following columns for each paper:

- Year, Conference, Filename, Title, Authors
- G1 (Focus), G2 (Attack Type), G3 (ML Type), G4 (Data Domain)
- G5 (Economics), G6 (Code Release), G7 (Real System Testing)
- T1 (Threat Model), T2 (Training Data Access)
- Q1 (Gradient Requirements), Q2 (Query Budget), Q3 (Computation)
- Gap indicator flags and Traditional Score

Benchmark meanings (aligned with the CSV fields used in the table):

- G1 Focus: atk (attack), def (defense), both.
- G2 Attack Type: Evasion, Poisoning, Privacy, Multiple.
- G3 ML Type: DL, Traditional, Both.
- G4 Data Domain: Images, Text, Audio, Malware, Other.
- G5 Economics mentioned: YES/NO.
- G6 Code released: YES/NO.
- G7 Real system testing: YES/NO.
- T1 Threat model: White-box, Gray-box, Black-box.
- T2 Training data access: Full, Partial, None.
- Q1 Requires gradients: YES/NO.
- Q2 Query budget: High (>1000), Low (<1000), None.
- Q3 Computation: High (GPU), Low (CPU).
- Traditional_Score (Gap Score 0–6): sum of six impractical-assumption flags (higher = less practical).

For full per-paper details (all 24 columns), please see the CSV file. Below is a compact, bordered summary table focused on the most critical taxonomy fields. Binary fields are rendered as checkmarks (yes) or blanks (no); threat/access levels are abbreviated (W/B/G/P, F/P, H/L).

Table 1: Compact summary of papers (key taxonomy fields).

Year	Venue	Paper (1st author)	G1	G2	G3	G4	G5	G6	G7	T1	T2	Grad	Qry	Comp	WB	Trad.
2022	ACM	Using	def	Poisoning	DL	Images	✓	✓	×	G	F	✓	H	L	0	3
2022	ACM	Reverse-Engineering	atk	Evasion	DL	Audio	✓	✓	×	B		×	H	L	0	2
2022	ACM	Tianshuo	def	Privacy	DL	Images	✓	✓	×	B		×	L	L	0	1
2022	ACM	CISPA	atk	Privacy	DL	Other	✓	×	×	B		×	L	L	0	2
2022	ACM	CISPA	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	School	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Physical	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Framework	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Florian	atk	Poisoning	DL	Images	✓	×	×	W	F	✓	H	L	1	5
2022	ACM	via	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	ExamplesforLeakage	def	Evasion	DL	Images	✓	✓	×	W	P	✓	H	L	1	4
2022	ACM	Harnessing	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Feature	atk	Privacy	DL	Other	✓	×	✓	B		×	L	L	0	1
2022	ACM	Imperial	atk	Privacy	Both	Other	✓	✓	×	B		×	H	L	0	2
2022	ACM	Learning	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	CISPA	atk	Evasion	DL	Text	✓	✓	×	B		×	H	L	0	2
2022	ACM	Amrita	def	Poisoning	DL	Images	✓	×	×	B		✓	H	L	0	4
2022	ACM	A	def	Evasion	Both	Text	✓	✓	×	B		×	L	L	0	1
2022	ACM	CISPA	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	ETH	def	Privacy	Both	Other	✓	×	×	B		×	L	L	0	2
2022	ACM	Hanqing	atk	Evasion	DL	Audio	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Stevens	atk	Privacy	DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Are	atk	Privacy	DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	When	atk	Evasion	DL	Audio	✓	✓	✓	B		×	H	H	0	1
2022	ACM	Learning	atk	Privacy	DL	Images	✓	✓	×	B		×	H	L	0	2
2022	ACM	Using	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	Neural	atk	Evasion	DL	Text	✓	✓	×	B	P	✓	H	H	0	3
2022	ACM	CISPA	atk	Privacy	DL	Images	✓	✓	×	B	P	×	L	L	0	1
2022	ACM	A	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	ACM	LPGNet:	def	Privacy	DL	Other	✓	✓	×	B		✓	H	L	0	3
2022	IEEE	Encoders	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	1Illinois	atk	Evasion	DL	Images	✓	✓	×	B		×	L	L	0	1
2022	IEEE	Graph	atk	Privacy	DL	Other	✓	✓	×	B	F	✓	H	L	0	3
2022	IEEE	Imperceptible	atk	Evasion	DL	Text	✓	✓	✓	B		×	L	L	0	0
2022	IEEE	Security	def		DL	Other	✓	✓	×	B		×		L	0	1
2022	IEEE	Jialuo	def		DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Yun	atk	Evasion	DL	Other	✓	✓	×	B		×	H	L	0	2
2022	IEEE	Production	atk	Poisoning	DL	Images	✓	×	×	W	P	✓	H	L	1	5
2022	IEEE	Classification	def		Both	Malware	✓	✓	×	B		×	L	L	0	1
2022	IEEE	in	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Markus	def	Privacy	Both	Other	✓	✓	×	B		×	L	L	0	1
2022	IEEE	Eugene	atk	Poisoning	DL	Text	✓	✓	×	B		✓	H	H	0	3
2022	IEEE	"Adversarial	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Attacks	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Weight	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	SoK:	def		Both	Other	✓	×	×	B		×		L	0	2
2022	IEEE	Borja	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Transformer	atk	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2022	IEEE	Property	atk	Poisoning	DL	Images	✓	✓	×	B	F	×	H	L	0	2
2022	IEEE	Yuhao	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2022	NDS	School	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	NDS	Federated	def	Poisoning	DL	Text	✓	×	×	W		×	L	L	1	3
2022	NDS	Property	atk	Privacy	DL	Images	✓	✓	×	B	F	✓	H	L	0	3
2022	NDS	RamBoAttack:	atk	Evasion	DL	Images	✓	✓	×	B		×	H	H	0	2
2022	NDS	Shengwei	atk	Privacy	DL	Images	✓	✓	✓	W	F	✓	H	L	1	3
2022	NDS	Robustness	both	Poisoning	DL	Images	✓	×	×	W	F	✓	H	L	1	5
2022	NDS	Ahmed	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	for	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Vocal	def	Evasion	DL	Audio	✓	✓	×	B		×		L	0	1
2022	USENIX	31st	atk	Privacy	Both	Other	✓	×	×	W	F	✓	H	L	1	5
2022	USENIX	August	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	August	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4

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2022	USENIX	sponsored	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	atk	Poisoning	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Attacks	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	sponsored	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	August	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	August	atk	Privacy	DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2022	USENIX	Novel	atk	Privacy	Both	Other	✓	✓	×	B		×	H	L	0	2
2022	USENIX	Technische	def		Both	Malware	✓	×	×	B		×		L	0	2
2022	USENIX	1CISPA	atk	Privacy	DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	CISPA	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Machine	def	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Virginia	atk	Poisoning	DL	Images	✓	✓	×	G	P	✓	L	L	0	2
2023	ACM	through	atk	Privacy	DL	Audio	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Stealing	atk	Evasion	DL	Text	✓	✓	×	B		×	H	L	0	2
2023	ACM	Stolen	both	Multiple	DL	Other	✓	✓	×	B	P	✓	H	L	0	3
2023	ACM	Information	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Deep	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Unforgeability	def		DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Stefano	def	Evasion	Traditional	Images	✓	✓	×	B		×	L	L	0	1
2023	ACM	Learning	atk	Poisoning	DL	Other	✓	✓	×	G	P	✓	H	L	0	3
2023	ACM	Secure	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Detection	atk	Evasion	DL	Malware	✓	✓	×	B		×	L	L	0	1
2023	ACM	Code	def		DL	Other	✓	✓	×	G		×	L	L	0	1
2023	ACM	and	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	through	atk	Privacy	DL	Other	✓	✓	×	B		✓	H	L	0	3
2023	ACM	Jingxuan	def		DL	Other	✓	✓	×	B		×	L	L	0	1
2023	ACM	and	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Evading	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	Changqing	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	ACM	AntiFake:	def	Evasion	DL	Audio	✓	×	✓	B		×	H	L	0	2
2023	ACM	against	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Effects	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Deep	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	arXiv:2112.04558v2	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Federated	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	SNAP:	atk	Poisoning	DL	Other	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Federated	def	Poisoning	DL	Images	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Information	atk	Privacy	DL	Text	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Covert	atk	Poisoning	DL	Images	✓	✓	×	B		✓	H	L	0	3
2023	IEEE	DepthFake	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Jiameng	atk	Evasion	DL	Text	✓	✓	×	B		×		L	0	1
2023	IEEE	Multi-party	def	Privacy	Both	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Andrew	def	Privacy	Both	Other	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Information	atk	Privacy	DL	Text	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Extraction	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2023	IEEE	Limin	atk	Poisoning	DL	Malware	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Learning	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Attacks	both	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Automatic	atk	Evasion	DL	Audio	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	yCybersecurity	def		DL	Images	✓	×	×	W		×	L	L	1	3
2023	IEEE	Ruijie	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	A	both	Privacy	DL	Images	✓	✓	×	B	P	×	H	L	0	2
2023	IEEE	SNAP:	atk	Poisoning	DL	Other	✓	×	×	B		×	H	L	0	3
2023	IEEE	Hong	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	BayBFed	def	Poisoning	DL	Images	✓	×	×	W	F	✓	H	L	1	5
2023	IEEE	Andre	atk	Evasion	DL	Audio	✓	✓	×	B	F	×	L	L	0	1
2023	IEEE	KASTEL	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	ELSA:	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Shengwei	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	Patch-agnostic	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	RAB:	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	ETH	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	and	def	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	through	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	IEEE	StyleFool:	atk	Evasion	DL	Images	✓	✓	×	B		×	L	L	0	1
2023	NDS	DNN	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Hadi	def	Evasion	DL	Audio	✓	✓	×	B		×	H	L	0	2

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2023	NDS	Yugeng	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Commercial	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Classifiers	def	Evasion	DL	Images	✓	✓	×	B		×	L	L	0	1
2023	NDS	Learning	def	Evasion	DL	Other	✓	✓	×	B		✓	H	L	0	3
2023	NDS	Jiayun	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Klim	atk	Evasion	DL	Other	✓	×	×	B		×	H	H	0	3
2023	NDS	Siyuan	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Universidad	def	Poisoning	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Deep	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Chunyi	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Robust	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	NDS	Physical	def	Evasion	Both	Other	✓	✓	✓	G	P	×	L	L	0	0
2023	NDS	*Technische	def	Privacy	Both	Text	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX		atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Detection	atk	Evasion	DL	Malware	✓	×	×	B		×	H	L	0	3
2023	USENIX	Scientific	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	of	def		DL	Other	✓	✓	×	B		×	L	L	0	1
2023	USENIX	for	atk	Poisoning	Both	Other	✓	×	×	W	F	✓	H	L	1	5
2023	USENIX	via	def	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	FREE	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	1CISPA	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Jonathan	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Vehicles	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Fairness	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Fairness	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Decompiling	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def		Both	Malware	✓	✓	×	B		×	L	L	0	1
2023	USENIX	Mazal	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def		DL	Malware	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Privacy	DL	Images	✓	✓	×	B	P	✓	H	L	0	3
2023	USENIX	is	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Federated	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	Kunal	atk	Evasion	DL	Malware	✓	✓	×	B	P	×	H	L	0	2
2023	USENIX	for	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	by	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	is	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	atk	Poisoning	DL	Text	✓	×	×	B		×	H	L	0	3
2023	USENIX	Universal	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2023	USENIX	is	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2023	USENIX	August	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	from	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Xueluan	def	Privacy	DL	Images	✓	×	×	B		×	L	L	0	2
2024	ACM	SafeEar:	def	Privacy	DL	Audio	✓	✓	×	B		×	L	L	0	1
2024	ACM	Against	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Multi-Agent	atk	Evasion	DL	Images	✓	✓	×	B		×	H	H	0	2
2024	ACM	Text-to-Image	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Autonomous	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Wei	atk	Poisoning	Both	Other	✓	✓	×	G	P	✓	H	L	0	3
2024	ACM	Byzantine-Robust	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Recognition	atk	Evasion	DL	Audio	✓	✓	✓	B		×		H	0	0
2024	ACM	Zijin	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	CSIRO's	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Items	atk	Poisoning	DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Applications	atk	Privacy	DL	Text	✓	✓	×	B		✓	H	H	0	3
2024	ACM	Attack	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Attachment	atk	Evasion	DL	Malware	✓	✓	✓	B		×	H	L	0	1
2024	ACM	Information	atk	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Alchemy:	def	Evasion	DL	Images	✓	✓	×	W		✓	H	L	1	4
2024	ACM	S2NeRF:	both	Privacy	DL	Images	✓	✓	×	B		✓	H	L	0	3
2024	ACM	Certified	atk	Poisoning	DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	ETH	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Jie	atk	Privacy	DL	Images	✓	✓	×	B	P	×	H	L	0	2
2024	ACM	Understanding	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	in	def	Evasion	DL	Malware	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	A	def	Privacy	DL	Images	✓	✓	×	B		×	H	L	0	2

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2024	ACM	The	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Fisher	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	into	atk	Privacy	DL	Text	✓	✓	×	B		×	H	L	0	2
2024	ACM	BadMerging:	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Optimization-based	atk	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Optimization-based	atk	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Model	atk	Privacy	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Shenzhen	atk	Privacy	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Information	def	Evasion	DL	Images	✓	✓	×	B		×		H	0	1
2024	ACM	Learning	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Blind	def		Both	Audio	×	×	×	B		×		L	0	3
2024	ACM	Information	atk	Privacy	DL	Text	✓	×	×	B		×	L	L	0	2
2024	ACM	Deepfake	def		DL	Audio	✓	✓	×	B		×		L	0	1
2024	ACM	Membership	atk	Privacy	DL	Images	✓	✓	×	B	P	×	L	L	0	1
2024	ACM	Membership	atk	Privacy	DL	Images	✓	✓	×	B	P	×	L	L	0	1
2024	ACM	Neural	atk	Privacy	DL	Images	✓	✓	×	B		×		L	0	1
2024	ACM		atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Fine-grained	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	from	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Institute	atk	Privacy	DL	Images	✓	✓	×	B		×	H	L	0	2
2024	ACM	Institute	atk	Privacy	DL	Images	✓	✓	×	B	F	×	H	L	0	2
2024	ACM	of	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	School	atk	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Examples:	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2024	ACM	Models	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	ACM	Learning	atk	Poisoning	DL	Images	✓	✓	×	B	P	×	L	L	0	1
2024	IEEE	Pattern	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Robin	def	Privacy	Both	Other	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Yuzheng	def	Privacy	Both	Images	✓	✓	×	B		×	L	L	0	1
2024	IEEE	Joshua	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Ziqi	def	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Linguistic	atk	Evasion	DL	Audio	✓	✓	✓	B		×	L	L	0	0
2024	IEEE	Language	atk	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	It's	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	into	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2024	IEEE	Test-Time	atk	Poisoning	DL	Images	✓	✓	×	G	P	✓	L	L	0	2
2024	IEEE	Xingshuo	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Mohammad	atk	Poisoning	DL	Images	✓	×	×	G	P	✓	H	L	0	4
2024	IEEE	Attacks	def	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2024	IEEE	Distribution	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Dropout	atk	Poisoning	DL	Images	✓	×	×	W	F	✓	H	L	1	5
2024	IEEE	GROV	def	Evasion	DL	Other	✓	✓	×	B		×	L	L	0	1
2024	IEEE	Mahmoud	atk	Evasion	DL	Other	✓	✓	×	B	P	×	L	L	0	1
2024	IEEE	Alec	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Accelerator	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	KASTEL	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	IEEE	Adversarial	def	Evasion	DL	Text	×	✓	×	W		×		L	1	3
2024	IEEE	2Services	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Large	def	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	DEMASQ:	def	Evasion	DL	Text	✓	×	×	B		✓	H	L	0	4
2024	NDS	Attacks	atk	Privacy	DL	Audio	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Poisoning	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	by	def	Privacy	DL	Images	✓	✓	×	B		×	L	L	0	1
2024	NDS	CamPro	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Ensemble	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Gelei	atk	Evasion	DL	Text	✓	×	×	B		×	H	H	0	3
2024	NDS	Watermarking	def	Evasion	DL	Audio	✓	✓	✓	W	F	×	L	L	1	1
2024	NDS	ICS	atk	Evasion	DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Federated	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Chengkun	def	Poisoning	DL	Text	✓	✓	×	B		✓	L	L	0	2
2024	NDS	Recognition	atk	Evasion	DL	Audio	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Transpose	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Transpose	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Inversion-based	def	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4

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Year	Venue	Paper (1st author)	G1	G2	G3	G4	G5	G6	G7	T1	T2	Grad	Qry	Comp	WB	Trad.
2024	NDS	Networks	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Browser	def	Privacy	Traditional	Other	✓	×	×	B		×		L	0	2
2024	NDS	Deep	def	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2024	NDS	Bang	def	Privacy	DL	Other	✓	✓	×	B		×	L	L	0	1
2024	NDS	Against	atk	Poisoning	DL	Images	✓	✓	×	B	P	✓	H	L	0	3
2024	NDS	Speaker	atk	Evasion	DL	Audio	✓	✓	✓	B		×		L	0	0
2024	NDS	Mitigating	def	Poisoning	DL	Images	✓	×	×	B		×		L	0	2
2024	NDS	for	def		DL	Malware	✓	✓	×	B		×	L	L	0	1
2024	NDS	in	both	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	NDS	Attacks	def	Poisoning	DL	Text	✓	✓	×	W	F	×	L	L	1	2
2024	NDS	Chaoxiang	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Tong	atk	Evasion	DL	Text	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Attacks	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Approach	def	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Qingzhao	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Zhenghang	def		Both	Other	✓	✓	×	B		×		L	0	1
2024	USENIX	Synthesis	def	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Measuring	def		DL	Audio	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Large	atk	Evasion	DL	Text	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Efficiency	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Efficiency	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Kathrin	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DNN-GP:	both	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	for	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DeepEclipse	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	DeepEclipse	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Between	atk	Multiple	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Red-Teaming	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Red-Teaming	atk	Evasion	DL	Images	✓	✓	×	B		×	H	L	0	2
2024	USENIX	Property	atk	Privacy	DL	Images	✓	✓	×	B	F	×	H	L	0	2
2024	USENIX	Transposed	def		DL	Images	✓	×	×	W	F	✓	H	L	1	5
2024	USENIX	in	def	Privacy	DL	Images	✓	✓	×	B		×	L	L	0	1
2024	USENIX	Shuaifan	def	Privacy	DL	Images	✓	✓	✓	W	F	✓	H	L	1	3
2024	USENIX	CISPA	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	CISPA	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Runtime	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Splitting	def	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Yixin	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Changjiang	atk	Poisoning	DL	Images	✓	✓	×	B	F	✓	H	L	0	3
2024	USENIX	Guangsheng	atk	Privacy	DL	Images	✓	×	×	W	F	✓	H	L	1	5
2024	USENIX	Bailey	def	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Meng	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	"Minxue	def	Privacy	DL	Images	✓	✓	×	W		✓	H	L	1	4
2024	USENIX	CISPA	atk	Privacy	DL	Images	✓	✓	×	B	P	×	H	H	0	2
2024	USENIX	False	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Shenchen	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	through	atk	Privacy	DL	Other	✓	×	×	B		✓	H	L	0	4
2024	USENIX	Shaofeng	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Mudjacking:	def	Poisoning	DL	Images	✓	×	×	B		✓	H	L	0	4
2024	USENIX	Randomised	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	33rd	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def		DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def	Privacy	Both	Text	✓	✓	×	B		×	L	L	0	1
2024	USENIX	August	atk	Privacy	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def		DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	is	def	Poisoning	DL	Images	✓	×	×	B		×		L	0	2
2024	USENIX	Attacks	def	Privacy	DL	Images	✓	✓	×	B		✓	H	L	0	3
2024	USENIX	Model	atk	Evasion	DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Zooming	both	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	atk	Privacy	DL	Text	✓	✓	×	B		×	L	L	0	1
2024	USENIX	is	def		DL	Text	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def		DL	Other	✓	✓	×	W	F	×	H	L	1	3
2024	USENIX	August	def		DL	Other	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	August	def	Evasion	DL	Images	✓	✓	×	B		×		L	0	1
2024	USENIX	August	def		Both	Malware	✓	×	×	B		×	L	L	0	2
2024	USENIX	Learning	def	Evasion	Both	Other	✓	✓	×	W	F	✓	H	L	1	4

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Year	Venue	Paper (1st author)	G1	G2	G3	G4	G5	G6	G7	T1	T2	Grad	Qry	Comp	WB	Trad.
2024	USENIX	Framework	def		DL	Text	✓	✓	×	B		×		L	0	1
2024	USENIX		def		DL	Images	✓	✓	×	B		×		L	0	1
2024	USENIX		atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2024	USENIX	Inference	def	Privacy	DL	Images	✓	×	×	G		✓	H	L	0	4
2025	ACM	1Technion	atk	Evasion	DL	Text	✓	✓	×	B		×	H	L	0	2
2025	ACM	in	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	arXiv:2501.05928v2	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Anti-Facial	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	BIFOLD	atk	Evasion	DL	Other	✓	✓	×	W	F	✓	H	H	1	4
2025	ACM	Targeted	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Language	atk	Poisoning	DL	Text	✓	✓	×	B		×	L	L	0	1
2025	ACM	Models	def	Evasion	DL	Images	✓	×	✓	G	P	✓	H	H	0	3
2025	ACM	Adversarial	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Construction	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Latent-based	atk	Evasion	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Milad	atk	Evasion	DL	Malware	✓	✓	✓	W	F	✓	H	L	1	3
2025	ACM	VillainNet:	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	ACM	Deep	def		DL	Malware	✓	✓	×	B		×		L	0	1
2025	USENIX	Jiachen	def	Evasion	DL	Images	✓	✓	✓	B		✓	H	H	0	2
2025	USENIX	Lingchen	def	Evasion	DL	Text	✓	✓	×	W	F	×	L	L	1	2
2025	USENIX	Revisiting	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	USENIX	Zhisheng	def	Privacy	DL	Audio	✓	✓	×	B		×	L	L	0	1
2025	USENIX	with	def	Evasion	DL	Other	✓	✓	×	W	F	×	H	L	1	3
2025	USENIX	Persistent	atk	Poisoning	DL	Images	✓	✓	×	W		✓	H	L	1	4
2025	USENIX	Benign	atk	Poisoning	DL	Images	✓	✓	×	W	F	✓	H	L	1	4
2025	USENIX	34th	def	Evasion	DL	Malware	✓	✓	×	B		✓	H	L	0	3